

Lesson Objective

Find the slope of a line on a grid using properties of slope triangles.

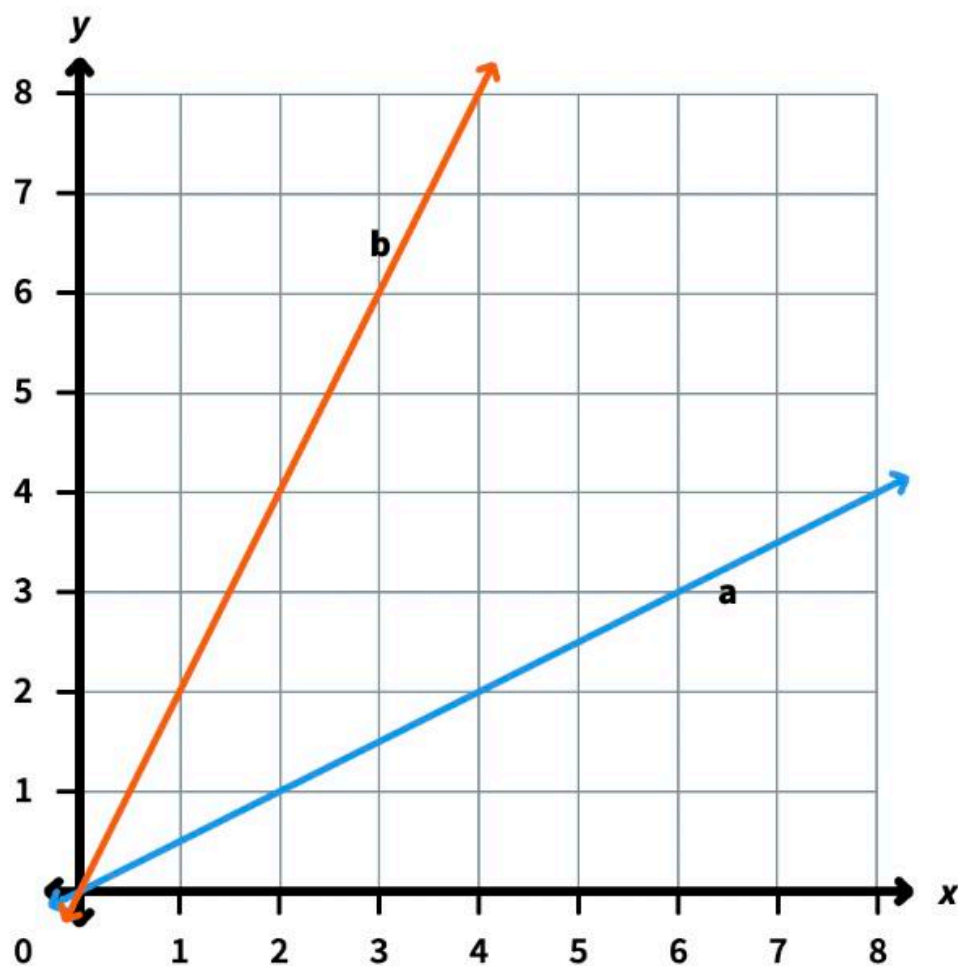
Materials

(S) Mini-Lesson Worksheet

PROBLEM

- 1 Identify which of two lines on a grid is steeper and connect steepness to the idea of slope as vertical distance divided by horizontal distance.

Two lines are graphed on the grid below. Which line is steeper, line *a* or line *b*?



Answer: Line b is steeper.

TEACHER GUIDANCE

Ask students which line is steeper. If it helps, tell students to imagine each line is a slide — the one they would go down faster is steeper.

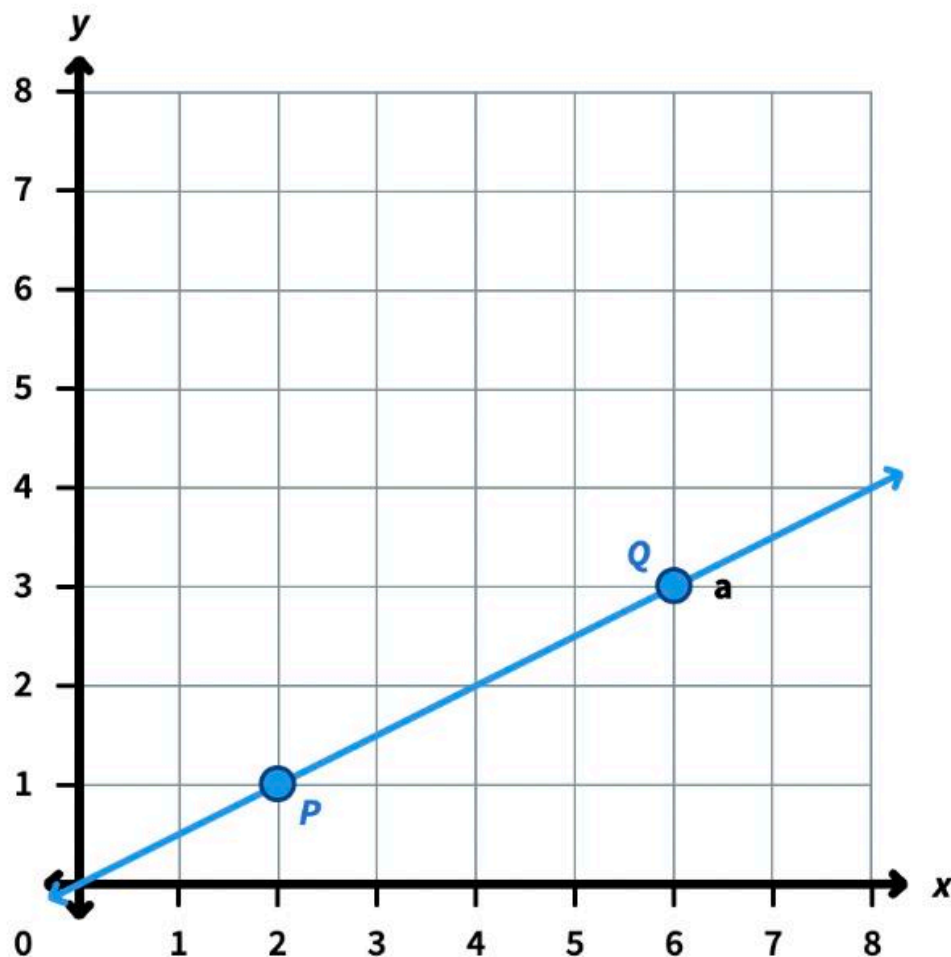
Tell students that slope is a number that describes the steepness of a line. Slope is the vertical distance divided by the horizontal distance between two points on the line. A steeper line has a greater slope.

Point out that in this lesson, students will learn how to find the slope of a line by picking two points on the grid and counting vertical and horizontal distances.

PROBLEM

- 2 Find the slope of line a by choosing two points at grid intersections and calculating the vertical distance divided by the horizontal distance.

Find the slope of line a .



Answer: The slope of line a is $2/4$, which equals $1/2$.

TEACHER GUIDANCE

Ask students to count the vertical distance from P to Q. Then ask students to count the horizontal distance from P to Q.

Have students record the slope as vertical distance divided by horizontal distance: $2/4$. Ask students to simplify: $2/4 = 1/2$.

Point out that a slope of $1/2$ means the line goes up 1 unit for every 2 units across.

- 3 Determine the slope of line b and compare slopes to describe the relative steepness of the two lines.

Find the slope of line b . Then explain how the slopes of line a and line b confirm which line is steeper.

Answer: The slope of line a is $2/4$, which equals $1/2$.

Have students find the slope of line b using points C and D. As students are ready, have them count the vertical and horizontal distances and calculate the slope independently.

Ask students to compare the slopes of line a and line b . Ask which slope is greater and how that matches what they observed about steepness in Problem 1. Confirm that line b has a slope of 2, which is greater than line a 's slope of $1/2$, so line b is steeper.

Monitor For

- Do students correctly identify the vertical distance and horizontal distance between two points?
- Are students counting grid squares accurately, or are they off by one?
- Do students write slope as vertical distance divided by horizontal distance, not the reverse?
- Can students connect a larger slope value to a steeper line?

Instructional Moves

- If students confuse vertical and horizontal distances, have them trace the path: "Go straight up first, then straight across" to form the slope triangle.
- Remind students that slope is always vertical distance divided by horizontal distance — "rise over run" — not the other way around.
- If students struggle to count distances accurately, have them mark each grid square as they count from one point to the other.
- When comparing slopes, ask: "Which slope number is greater? That line is steeper. Why does that make sense?"

Reflection Questions

1. How did you use the grid to find the slope of a line?

Answer: I picked two points where the line crosses grid intersections, counted the vertical distance and the horizontal distance between them, and divided vertical distance by horizontal distance.

2. What does a slope of $\frac{1}{2}$ tell you about how the line looks compared to a slope of 2?

Answer: A slope of $\frac{1}{2}$ means the line only goes up 1 unit for every 2 units across, so it is less steep. A slope of 2 means the line goes up 2 units for every 1 unit across, so it is much steeper.

3. How does comparing slope values help you decide which line is steeper?

Answer: The line with the greater slope is steeper because it has a greater vertical distance for the same horizontal distance.

Mini Lesson Sample Script

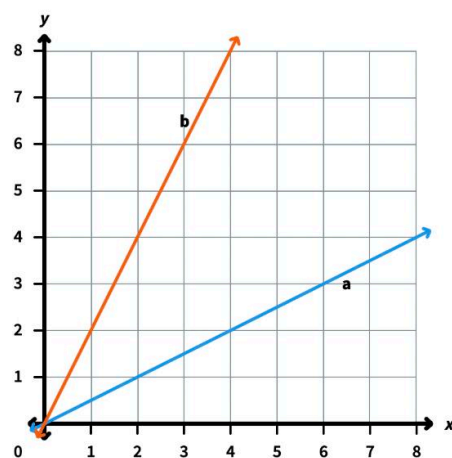
Materials

(S) Activity 1 Worksheet

PROBLEM

- 1 Identify which of two lines on a grid is steeper and connect steepness to the idea of slope as vertical distance divided by horizontal distance.**

Two lines are graphed on the grid below. Which line is steeper, line *a* or line *b*?



Answer: Line b is steeper.

SAMPLE SCRIPT

T: Look at the two lines on your worksheet. Imagine each line is a slide at a playground. If you slid down both, which one would you go faster on?

S: Line b. It's way more tilted.

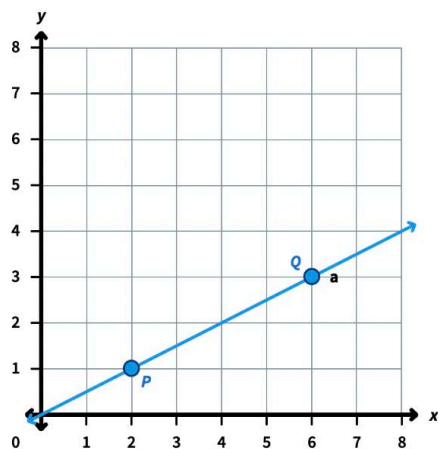
T: Right, line b is steeper. In math, we have a number that describes how steep a line is. It's called slope. Slope is the vertical distance divided by the horizontal distance between two points on a line. A steeper line has a greater slope. Today we're going to find slopes by picking two points on a line and counting distances on the grid.

PROBLEM

SAMPLE SCRIPT

- 2 Find the slope of line a by choosing two points at grid intersections and calculating the vertical distance divided by the horizontal distance.**

Find the slope of line a .



Answer: The slope of line a is $\frac{2}{4}$, which equals $\frac{1}{2}$.

T: Look at line a. I've marked two points, P and Q, where the line crosses grid intersections. Starting at point P, count straight up to the level of point Q. How many units is that?

S: 2 units.

T: That's the vertical distance. Write that down. Now, from there, count straight across to point Q. How many units?

S: 4 units.

T: That's the horizontal distance. So what is the slope? Remember, slope is vertical distance divided by horizontal distance.

S: 2 divided by 4. That's $\frac{2}{4}$.

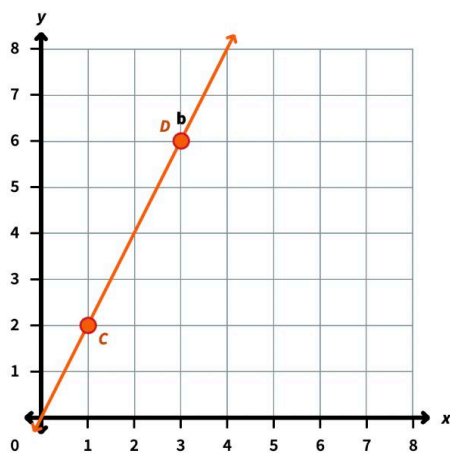
T: Can we simplify that?

S: $\frac{1}{2}$.

T: So the slope of line a is $\frac{1}{2}$. That means for every 2 units you go across, the line goes up 1 unit. Write that on your worksheet.

- 3 Determine the slope of line b and compare slopes to describe the relative steepness of the two lines.**

Find the slope of line b . Then explain how the slopes of line a and line b confirm which line is steeper.



Answer: The slope of line b is $\frac{4}{2}$, which equals 2. Line b is steeper because its slope of 2

T: Now find the slope of line b using points C and D. Count the vertical distance and horizontal distance, then calculate the slope. Go ahead.

S: (Work.) The vertical distance from C to D is 4 units. The horizontal distance is 2 units. So the slope is 4 divided by 2, which is 2.

T: Good. So the slope of line b is 2. Now, think back to Problem 1. We said line b was steeper. Look at the two slopes: line a has a slope of $\frac{1}{2}$, and line b has a slope of 2. Which slope is greater?

S: 2 is greater than $\frac{1}{2}$.

T: And that matches what we saw — line b is steeper. A larger slope means the line goes up more for the same distance across, so it's steeper.